Trademark Matcher

User Manual and Comparison Filter Guide

What every setting means, how the similarity percentage is calculated, and how to tune the comparison for your catalogue.

Covers: catalogues, source image, result filter, extraction settings, image preparation, variations, the eight metrics, saved filter settings and search history.

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1. How the app works

Trademark Matcher answers one question: **which trademarks in my PDF catalogue look like this one?** You give it a source image and pick a catalogue. It measures how alike the source is to every catalogue trademark, converts the result to a percentage, and lists only those at or above the minimum similarity you set. Each result shows the source and the match side by side with the percentage between them.

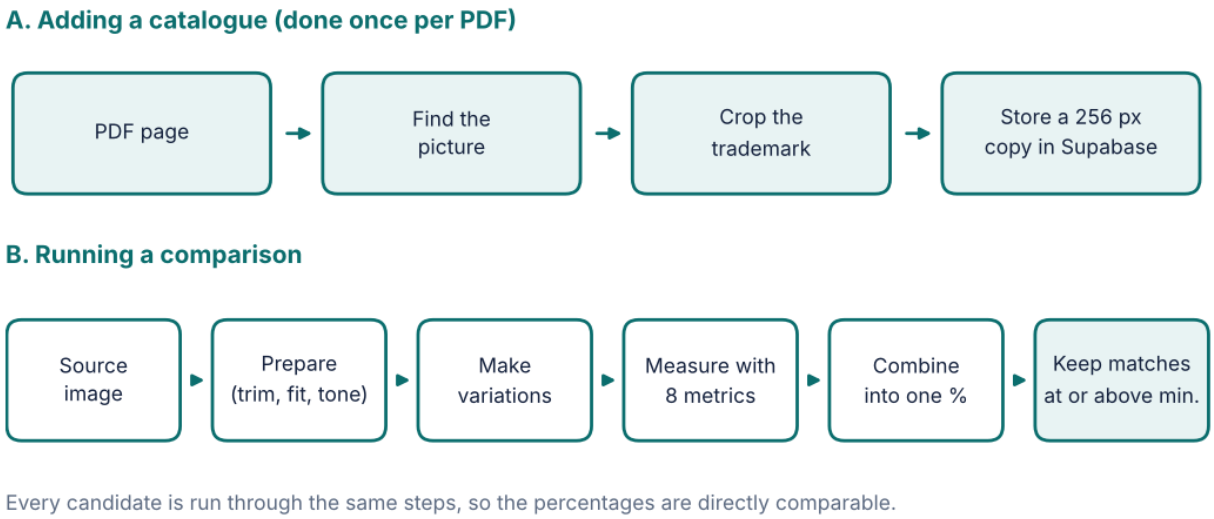


Figure 1. Adding a catalogue (A) and running a comparison (B).

The comparison does not use a single test. It runs **eight different metrics**, each looking at the marks in a different way (overall layout, outlines, silhouette, colour and so on), then combines them using the weights you choose. This lets you tune the app: for example, ignore colour to find the same design in other colours, or demand that every test agrees to find only near-copies.

What it can and cannot tell you

- It measures **visual** similarity of the images only.
- It does not read text, compare sounds, goods and services or legal status. A high score is a reason to look closer, not a legal finding of conflict or infringement, and a low score does not clear a mark.
- All image processing happens in your browser. Only the catalogue copies, search history and saved settings are stored in your Supabase project, and only you can see them.

2. Quick start

Step	What to do
1. Sign in	Create an account with email and password, or use the email link. Confirm your email if the app asks you to.
2. Add a catalogue	Open the **Catalogues** tab. Choose the PDF (one trademark image per page, plus unrelated text), give it a name and press **Upload and extract**. The app finds the picture on each page, crops it and stores it. This is done once per PDF.
3. Add the source	Open **Compare**. Tap the dashed box to choose your source trademark. PNG, JPG, WebP, GIF, BMP, SVG and AVIF work. You can also drop or paste an image.
4. Choose the catalogue	Pick it in **Compare against**.
5. Set the filters	Set the **Minimum similarity**. Tap the ? beside any option for a short explanation. Or load one of your saved settings.

Step	What to do
6. Compare	Press **Compare**. Matches appear best first. Tick **Save this search to History** to keep the results.

> **Tip:** start with the default settings and a minimum of 60%. Look at the first few results, then adjust. Section 7 gives ready-made recipes.

3. Reading the results

Each result card shows the **source** on the left, the **catalogue trademark** on the right (with its PDF page number), and the **overall similarity** in the middle. Under the images, one bar per active metric shows how that individual test scored. The last line names the source variation that gave the best score, such as “rotated 90°” or “as uploaded”.

Percentage	Colour	Rough meaning
85% and above	Green	Very close. Same or nearly the same artwork. Review first.
70% to 84%	Amber	Clearly related. Shared shape, layout or outline. Worth a look.
Below 70%	Blue	Weaker resemblance. Shown only if your minimum is set lower.

These bands are a guide, not rules. How a given percentage feels depends on your weights: a stricter setup gives lower numbers for the same pair of marks. Compare scores only within the same filter settings.

If nothing reaches your minimum, the app tells you the closest score it found, so you know how far to lower the minimum.

The per-metric bars are the best diagnostic. If the overall score is high only because Color is 100%, the shapes may be unrelated. If Shape and Edges are high but Color is low, it is probably the same design in different colours.

4. Filter reference

Every filter has a ? button in the app that shows a short version of the text below. Settings are grouped as they appear on screen. Defaults are shown in the tables.

4.1 Result filter

Setting	Default	Meaning and advice
Minimum similarity (%)	60	Only matches at or above this overall percentage are listed. Raise it (75 to 90) to see only close look-alikes. Lower it (40 to 55) for wide screening. Because the overall score depends on your weights, re-check this value when you change them.
Maximum results shown	50	The most results listed, best first. History saves the same number. Matches beyond the limit are counted but not shown.

4.2 Extraction settings (Catalogues tab)

These apply when a PDF is uploaded and cannot be changed afterwards. To try different settings, upload the PDF again as a new catalogue.

Setting	Default	Meaning and advice
Take from each page	Largest embedded image	**Largest:** the biggest picture on the page is the trademark. Best for one mark per page with unrelated text. **All:** every picture becomes a separate candidate, for pages with several marks. **Whole page:** ignores embedded pictures and uses the whole page, for scanned catalogues where each page is one flat image.
Page render quality	High (2×)	How sharply each page is drawn before cropping. Higher is sharper but slower and uses more memory. Use 3× only for tiny marks; use 1× for very large PDFs on slow devices.
Ignore images smaller than (px)	60	Skips small pictures such as icons or stamps. Raise it if page decorations are being picked up. Lower it if real marks are tiny.
First page / Last page	1 / 0 (all)	Reads only part of the PDF. Last page 0 means to the end. Useful for testing on a few pages first.

If a page has no embedded picture, the whole page is used and the app reports how many pages did this. Many of them usually means the PDF is scanned: switch to **Whole page**.

4.3 Preparing images

Before measuring, both images go through the same preparation so that size, position and background do not distort the comparison.



Figure 2. Trimming margins and fitting into a square. The last two panels show the two resize methods on a wide mark.

Setting	Default	Meaning and advice
Comparison resolution (px)	64	Size each image is shrunk to. 32 is fastest and coarsest. 128 keeps fine detail but is about four times slower than 64. Catalogue images are stored at 256 px, so there is no gain above 128. 64 suits most logos; use 96 or 128 for marks with thin lines.
Resize method	Keep proportions	**Keep proportions** never distorts the mark; spare space is filled with the background colour. **Stretch to square** forces the mark to fill the square, so wide and tall versions of one design look more alike but real shapes are bent.
Trim background margins	On	Removes empty border so the position and size of the mark on its page do not matter. Turn off only if you want position within the frame to count.
Background tolerance (%)	12	How different from the background a pixel must be to count as part of the mark when trimming. Raise (20 to 30) for noisy or off-white scans so specks are ignored. Lower (3 to 8) for very faint marks. Has no effect if trimming is off.
Normalise contrast	Off	Stretches brightness so the darkest pixel becomes black and the lightest white. Helps compare a faded scan against a crisp digital file. Leave off for clean images.
Shape threshold (0 = automatic)	0	The brightness level separating mark from background for the Shape metric only. 0 chooses it automatically for each image. Set a value only if shapes look wrong. Lower values count only the darkest areas as the mark.

4.4 Variations of the source to test

When variations are on, the app compares extra transformed copies of the **source** against each catalogue trademark and keeps the **best** score. This finds marks that are the same design but turned, reversed or inverted.



Figure 3. Variations of the source that can be tested.

Setting	Default	Meaning and advice
Rotations to try	None	Every 90° tests four orientations. Every 45°, 30° or 15° tests 8, 12 or 24 orientations. More angles find more turned marks but multiply the time. Start with every 90°.
Also try mirrored	Off	Adds left-to-right mirror copies. Doubles the work.
Also try inverted colors	Off	Adds a light-on-dark version. Finds a white logo on black against a black logo on white. Doubles the work.

Options multiply. Rotations every 90° plus mirrored plus inverted tests 4 x 2 x 2 = 16 variations of the source against every catalogue mark. Every 15° with both options is 96 variations; use it only on small catalogues.
A higher chance of finding a match also means a higher chance of false positives. Because only the best variation counts, turning variations on can raise scores for unrelated marks.

5. The eight metrics

Each metric gives a score from 0% (nothing alike) to 100% (identical by that test). The **weight** slider (0 to 10) sets how much that metric counts in the overall result. A weight of 0 switches the metric off and hides its bar.

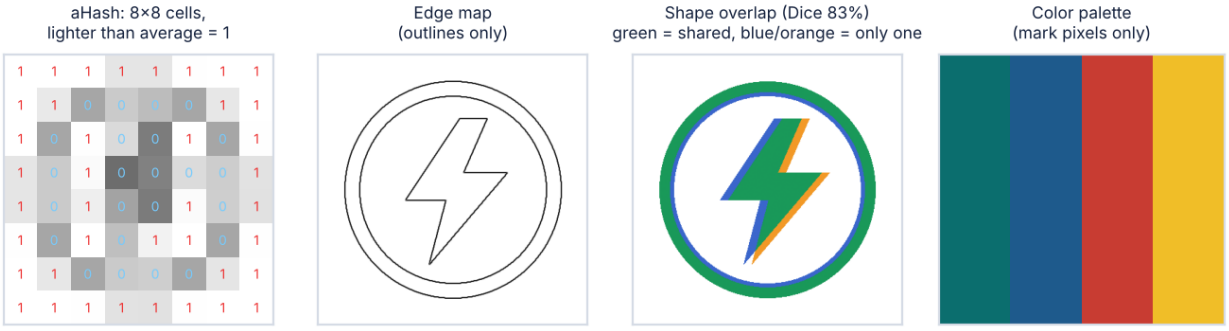


Figure 4. Four of the ideas behind the metrics, shown on two similar marks.

Hash metrics (pHash, dHash, aHash)

A hash turns an image into a short fingerprint of 64 yes/no values. Two images are compared by counting how many values differ. About half of the values differ between unrelated images, so the app rescales hash scores so that **50% of values different counts as 0%** and identical counts as 100%. Hashes are fast and tolerant of small changes in size, sharpness and colour.

Metric	How it works	Strong at	Weak at
pHash overall structure (weight 2)	Shrinks to 32 x 32 grey, applies a frequency transform (DCT) and keeps the 63 lowest-frequency values. Each is 1 if above the median.	The broad light and dark layout of the design. Resists blur, resizing and colour shifts.	Fine detail and thin lines. Marks with the same broad layout but different details can score high.
dHash gradients (weight 1)	Shrinks to 9 x 8 and records, for each cell, whether it is brighter than its right-hand neighbour.	Direction and position of edges and transitions.	Fine texture. Sensitive to rotation and flipping.
aHash brightness layout (weight 0.5)	Shrinks to 8 x 8 and records which cells are lighter than the average.	Very quick rough check of the overall light and dark balance.	Easily fooled. Keep its weight low.

Pixel and structure metrics

Metric	How it works	Strong at	Weak at
SSIM local similarity (weight 2)	Splits the image into 8 x 8 pixel blocks and compares brightness, contrast and pattern in each, in the way human vision does. Blocks empty in both images are skipped so blank background cannot inflate the score. Result is the average over the remaining blocks.	Near-identical artwork, small local differences, changes in line weight.	Shifts and rotations. A slightly moved mark scores much lower.
Pixel correlation (weight 1)	Measures whether bright and dark pixels rise and fall together across the two images (Pearson correlation). Negative values count as 0.	Close copies at the same position and scale.	Position. Strict about alignment, so it punishes any shift.

Metric	How it works	Strong at	Weak at
Edge / outline (weight 1.5)	Finds the strokes and borders in each image (Sobel filter), blurs them slightly, and measures how well the two outline maps agree (cosine similarity).	Line-art logos. The same design with different fill or colour.	Marks that differ mainly by solid fills; very thick solid shapes have few edges.

Shape and colour metrics

Metric	How it works	Strong at	Weak at
Shape overlap (weight 2)	Turns each mark into a solid black-and-white silhouette using the shape threshold. The smaller side is treated as the mark, so light-on-dark and dark-on-light behave alike. Score is the Dice coefficient: 2 x shared area / (area of mark A + area of mark B).	Bold, simple marks and silhouettes, regardless of colour or inner detail.	Marks that differ only in inner detail. Thin line art makes poor silhouettes.
Color palette (weight 0.5)	Sorts the mark pixels (background excluded) into 64 colour bins and measures how much the two colour distributions overlap.	Telling apart marks of the same shape in different colours.	Says nothing about shape. Black-and-white marks all look alike. Set weight 0 to ignore colour.

> **Note:** the structure, outline, shape and colour metrics are not rescaled like the hashes. Two unrelated marks with lots of empty background can still score a fair amount on them. This is why a minimum around 60% is a sensible starting point, and why combining several metrics works better than any one alone.

5.1 Combine metrics by

This chooses how the individual scores become one percentage. Only metrics with a weight above 0 take part.

Mode	Result	Use it for
Weighted average	Sum of (score x weight) divided by the sum of weights.	Balanced, general screening. The default.
Strictest metric	The lowest score among active metrics.	Finding near-copies. A match must pass every active test, so use few, well-chosen metrics.
Best metric	The highest score among active metrics.	Wide screening. One strong resemblance is enough, so expect many false positives.

Worked example

Metric	Score	Weight	Score x weight
pHash	88%	2	1.760
dHash	75%	1	0.750
aHash	69%	0.5	0.345
SSIM	81%	2	1.620
Pixels	74%	1	0.740
Edges	79%	1.5	1.185
Shape	90%	2	1.800
Color	60%	0.5	0.300
Total		**10.5**	**8.500**

Weighted average: 8.500 / 10.5 = **81.0%**. Strictest metric: **60%** (Color). Best metric: **90%** (Shape). The same pair of marks gives three very different overall percentages, which is why the mode and your minimum similarity must be chosen together.

6. Saved filter settings

Once you have a combination that works, save it under a name and reuse it later. Saved settings belong to your account and are available on any device you sign in on.

What is saved

Everything in the **Result filter**, **Preparing images**, **Variations** and **Metric weights** sections: minimum similarity, maximum results, resolution, resize method, trimming and tolerance, contrast, shape threshold, rotations, mirrored, inverted, combine mode and all eight weights. Extraction settings are not included because they belong to each catalogue.

How to use them

To do this	Do this
Save the current filters	In **Compare**, open **Saved filter settings**. Type a name in the name box, for example “Strict near-copy”, and press **Save**.
Update a saved setting	Load it, change the filters, keep the same name and press **Save**. The message says “Updated”.
Load a saved setting	Choose it in the drop-down. All controls change at once.
Delete a saved setting	Choose it in the drop-down and press **Delete**. This cannot be undone.
Go back to defaults	Press **Reset**, or choose “Default settings” in the drop-down.

The setting you used last is selected automatically the next time you open the app on the same browser.

Search history

With **Save this search to History** ticked, each comparison stores the source image, the filter settings used and the matches found. Open the **History** tab to reopen any search exactly as it was, even after you change your saved settings, or to delete it. Searches that found no matches are not saved.

Reviewing matches

Saved matches can be reviewed. Each result card has a **status** drop-down and a **note** box. Changes save automatically.

Status	Use it for
New	Not looked at yet. Every match starts here.
Cleared	Reviewed and judged not a concern.
Watch	Related enough to monitor, but no action yet.
Conflict	Needs action. Shown with a red edge on the card.

In **History**, use **Show matches with status** to list only one status, for example all Conflicts. The line beneath it counts matches in each status.

Exporting and printing

Above the results, **Download CSV** saves a spreadsheet with rank, page, similarity, every metric score, best variation, status, note, source, catalogue and date. It opens in Excel, Numbers or Google Sheets. **Print / save as PDF** opens your browser print dialog with a clean report: a heading with the source, catalogue, date and filter settings, followed by the match cards with their status and notes. Choose “Save as PDF” as the printer to keep a file. Both buttons work on the current results, including a filtered History view.

7. Recommended starting recipes

Use these as starting points, then adjust while watching the per-metric bars. Items not listed stay at their defaults. “Min” is the minimum similarity.

Goal	Suggested settings
Broad screening Find anything that might be related	Min 50 to 55. Combine by weighted average. Variations: rotations every 90°. Keep the default weights.
Near-copies only Same artwork, small changes	Min 80. Combine by strictest metric. Weights: pHash 2, SSIM 2, Edges 2, Shape 2; set dHash, aHash, Pixels and Color to 0.
Same design, any colours	Weights: Color 0, aHash 0. Raise Edges to 3 and Shape to 3. Turn on inverted colors if black-and-white versions exist.
Shape and silhouette Simple bold marks	Weights: Shape 6, pHash 3, Edges 1; others 0. Shape threshold 0 (automatic). Resolution 64.
Line art and fine detail	Resolution 128. Weights: Edges 4, SSIM 3, pHash 2, Shape 1; others 0. Lower background tolerance to 6.
Turned or reversed marks	Rotations every 45° (or 90° first), mirrored on. Keep weighted average and raise the minimum 5 points to offset extra false positives.
Faded or noisy scans	Normalise contrast on. Background tolerance 20 to 30. Resolution 64. Reduce Pixels weight to 0.5.

8. Troubleshooting

Problem	What to try
The file picker does not open	Tap directly on the dashed box. If you are using an in-app viewer, open the site in your phone’s normal browser. You can also drag a file onto the box or paste an image.
Source image will not load	Convert it to PNG or JPG. TIFF is not supported by browsers.
Upload fails or stops	The free Supabase plan limits files to 50 MB. Split a larger PDF. Check your connection and try again; a failed upload is cleaned up automatically.
Wrong picture was taken from a page	Raise **Ignore images smaller than**, or use **All embedded images** and compare results. Upload again as a new catalogue.
“No embedded image” message	The PDF pages are probably scans. Re-upload with **Whole page**.
Too many weak matches	Raise the minimum, switch to strictest metric, lower weights of aHash, Pixels and Color, or turn variations off.
Obvious look-alike missing	Lower the minimum. Turn on rotations, mirrored or inverted. Check the Preparing images options, especially trimming and tolerance.
Comparison is slow	Lower the resolution to 32 or 64, reduce variations, or compare against a smaller catalogue. The first search in a session also downloads the catalogue images.
Sign-in link or confirmation fails	Check that your site address is listed under Authentication, URL Configuration in Supabase.

9. Glossary

Term	Meaning
Catalogue	A PDF you have uploaded, stored as one cropped image per trademark.
Source trademark	The image you are searching for.
Variation	A rotated, mirrored or inverted copy of the source tested alongside the original.
Metric	One way of measuring similarity, scored 0% to 100%.

Term	Meaning
Weight	How much a metric counts towards the overall score. 0 switches it off.
Hash	A short fingerprint of an image, compared by counting differing values.
Silhouette	The mark reduced to solid black on white, ignoring colour and inner detail.
Dice coefficient	Overlap measure: twice the shared area divided by the combined area of both shapes.
Threshold	A brightness level used to decide which pixels belong to the mark.
Preset / saved setting	A named set of comparison filters stored in your account.

Appendix: default values

Group	Defaults
Result filter	Minimum 60%. Maximum results 50.
Extraction	Largest embedded image. Quality 2x. Ignore under 60 px. All pages.
Preparing images	Resolution 64. Keep proportions. Trim on. Tolerance 12%. Normalise off. Shape threshold automatic.
Variations	No rotation. Mirrored off. Inverted off.
Combine	Weighted average.
Weights	pHash 2, dHash 1, aHash 0.5, SSIM 2, Pixels 1, Edges 1.5, Shape 2, Color 0.5.